KERALA WATER AUTHORITY

JAL JEEVAN MISSION (JJM)-JJM - WSS TO NEDUMKANDAM, PAMPADUMPARA, UPPUTHARA ELAPPARA (PART) AND ARAKULAM (PART) PANCHAYATHS IN IDUKKI DISTRICT -- SUPPLYING AND LAYING OF CWPM, SPH AND STEEL TANK AT NEDUMKANDAM

Re-E -TENDER No: 17/2024-25/SE/PHC/IDK/2951

DUE DATE: 29.11.2024 up to **16.00** Hrs.

VOLUME – II

SCOPE OF WORK

OFFICE OF THE SUPERINTENDING ENGINEER

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NAME OF WORK: JAL JEEVAN MISSION (JJM)-JJM - WSS TO NEDUMKANDAM, PAMPADUMPARA, UPPUTHARA, ELAPPARA (PART) AND ARAKULAM (PART) PANCHAYATHS IN IDUKKI DISTRICT - SUPPLYING AND LAYING OF CWPM, SPH AND STEEL TANK AT NEDUMKANDAM

1. GENERAL INFORMATION

The Project is aimed for providing potable drinking water to the households in Nedumkandam, Pampadumpara, Upputhara, Elappara (part) and Arakulam (part) Panchayaths in Idukki District of Kerala State. The scheme area is situated at the western part of Idukki district of Kerala. Design period of the scheme is taken as 30 years from 2023 A.D. with a per capita demand of 100 lpcd.

The proposed project includes following packages:

1. Supplying and Laying Distribution System and Providing FHTCs in Upputhara, Elappara(part) and Arakkulam(part) Panchayaths in Idukki district.
2. Supply and laying distribution system and Providing FHTCs in Pampadumpara Panchayath.
3. Supply and laying distribution system and Providing FHTCs in Nedumkandam Panchayath.
4. Supplying and laying of CWPM from WTP to Kalyanathandu top including construction of Sump cum pump house and 11 LL GLSR.
5. Providing and Laying CWGM from Kalyanathandu to Nedumkandam SPH and construction of SPH.
6. Providing and Laying CWGMs, CWPM, construction of SPH, 6.5 LL GLSR in Upputhara and 0.65 LL GLSR in Ambalamedu.
7. Providing and laying CWPMs & CWGM and Construction of SPH & GLSR in Pampadumpara.
8. Supplying and laying of CWPM, SPH and Steel tank at Nedumkandam.
9. Supply, erection and commissioning of Transformer and pump sets for all zones. Power connection and electrification.

10. Construction of new 35 MLD capacity water treatment plant.

11. Supply and erection of Floating Pumphouse.

12. Soil Investigation at Anchuruli WTP for Two Clariflocculators.

13. Soil Investigation at Anchuruli WTP for Aerator and Flash Mixer.

14. Soil Investigation at Anchuruli WTP for Filter hose and Settling tank

15. Soil investigation at Pampadumpara Estate Top for 5LL GLSR

16. Soil investigation at Pampadumpara Estate Top for 5LL GLSR- Site 1

This tender is invited for the package - "Providing and laying CWPMs & CWGM and Construction of SPH & GLSR in Pampadumpara".

The scope of work includes supply and laying of CWPM, Road restoration etc. at Nedumkandam panchayath, construction of 2.9 LL SPH at Pushpakandam, 3 LL Sump cum Pump House at Bedumettu, and steel tanks at Kailasapara, Shoolapara and handing over to Kerala Water Authority. The detailed design and drawing should be submitted within 30 days from the date of agreement.

The major components of proposed project are supply and laying of CWPM, road restoration and construction of 2.9 LL SPH at Pushpakandam, 3 LL Sump cum Pump House at Bedumettu, and steel tanks at Kailasapara and Shoolapara.

2. SCOPE OF WORK

This tender is invited as item rate tender for the following works.

i. Supply and laying 150mm, 200mm, 250mm DI K9 pipe lines for CWPMs.

ii. Road restoration work

iii. Construction of 2.9 LL Sump cum Pump House at Pushpakandam, 3LL Sump cum Pumphouse at Bedumettu and steel tanks at Kailasapara and Shoolapara.

iv. Road restoration work:

a) Restoration of earthen berm.
Restore the earthen berm after pipe laying using granular sub base and compaction to required level.

b) Restoration of concrete portion
Restore the concrete portion of road after pipe laying after providing sub base and base using GSB including compaction and PCC/RCC 1:3:6 and M20.

c) Restoration of bituminous surface
Restore the bituminous portion of road after pipe laying after providing sub base and base, surface course using GSB, WMM, BM and BC with necessary compaction.

2.1 DETAILED SCOPE OF WORK

2.1.1 Design and construction of Steel Tanks at Kailasapara, Shoolapara and SPHs at Pushpakandam and Bedumettu

Construction of Ground Level Service Reservoir (G.L.S.R) including construction, testing and commissioning of Ground level RCC reservoirs having effective storage capacity as mentioned in the table below along with all associated works viz: supply, installation, of DI/C.I D/F pipes, specials, valves etc. for Inlet/ Outlet/ Overflow/ Scour/ Interconnection with Distribution system. The thickness of side wall shall be a minimum of 250 mm and bottom slab shall be a minimum of 200 mm. The thickness of cover slab of tank shall be a minimum of 200 mm with clear cover as per the exposed condition “Severe”

specified in the Code IS 456:2000 (with latest amendments).

Sl No	GLSR	Capacity (in lakh Lts)	Pumphouse dimension in m2	Wall casting pipe details
1	Sump Cum Pump House at Pushpakandam	2.9	18.762	Suitable size as directed
2	Sump Cum Pump House at Bedumettu	3	11.28	Suitable size as as directed
3	Steel tank at Kailasapara	1.5	-	Suitable size as directed
4	Steel tank at Shoolapara	3	-	Suitable size as directed

The work abstracted in clause 2.1 above includes the completion of works specified from i to xxxiii below as per the directions of site Engineer.

i. Preparation of layout plan for the site establishment and work program including soil investigation.
ii. Clearing jungle including uprooting of rank vegetation, grass, brush wood, trees and saplings of girth up to 30 cm measured at a height of 1 m above ground level and removal of rubbish up to a distance of 50 m outside the periphery of the area cleared.
iii. Earth work in excavation by mechanical means (Hydraulic excavator) /manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift up to 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m.All kinds of soil
iv. Earth work in excavation by mechanical means (Hydraulic excavator)/ manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including disposal of excavated earth, lead up to 50 m and lift up to 1.5 m, disposed earth to be levelled and neatly dressed. Hard rock (requiring blasting)
v. Earth work in surface excavation not exceeding 30 cm in depth but exceeding 1.5m in width as well as 10 sqm on plan including disposal of excavated earth up to 50 m and lift up to 1.5 m, disposed soil to be levelled and neatly dressed - All Kinds of

soil.

vi. Providing and laying in position cement concrete of specified grade - All work up to plinth level 1:2:4 (cement: 2 coarse sand (zone-III): 4 graded stone aggregate 20 mm nominal size).

vii. Providing and laying in position specified grade of reinforced cement concrete- All work up to plinth level:1:1:5:3 (1 cement 1.5 coarse sand :3 graded stone aggregate 20 mm nominal size.

viii. Supply of Sand including loading, unloading, transportation and other incidental charges as per the direction of departmental officers.

ix. Supply, installation and commissioning of pre-engineered, pre-fabricated, factory manufactured steel storage Water Tank having a capacity of 150000 L.

x. Supply, installation and commissioning of a pre-engineered, prefabricated, factory manufactured steel storage Water Tank having a capacity of 300000 L.

xi. Providing and laying in position machine batched and machine mixed design mix M-25 grade cement concrete for reinforced cement concrete works using cement content as per approved design mix, including pumping of concrete to site, also including admixtures in recommended proportions as per IS: 9103 to accelerate, retard setting of concrete, improve workability without impairing strength and durability as per direction of Engineer - in-charge. Note: - Cement content considered in this item is @ 330 kg/ cum. Excess or less cement used as per design mix is payable or recoverable separately.

xii. Providing richer mixes (M-30) at all floor levels.

xiii. Providing and applying integral crystalline slurry of hydrophilic in nature for waterproofing treatment to the RCC structures like retaining walls of the basement, water tanks, roof slabs, podiums, reservoir, sewage & water treatment plant, tunnels / subway and bridge deck etc., prepared by mixing in the ratio of 5 : 2 (5 parts integral crystalline slurry : 2 parts water) for vertical surfaces and 3 : 1 (3 parts integral crystalline slurry : 1 part water) for horizontal surfaces and applying the same from negative (internal) side with the help of synthetic fiber brush. The material shall meet the requirements as specified in ACI-212-3R-2010 i.e by reducing permeability of concrete by more than 90% compared with control concrete as per DIN 1048 and resistant to 16 bar hydrostatic pressure on negative side. The crystalline slurry shall be capable of self-healing of cracks up to a width of 0.50mm. The work shall be carried out all complete as per specification and the direction of the engineer in charge. The product performance shall carry guarantee for 10 years against any

leakage. For vertical surface two coats @0.70 kg per sqm and one coat @1.10 kg per sqm for horizontal surface.

xiv. Centering and shuttering including strutting, etc. and removal of form for: walls (any thickness), attached pilasters, butteresses, plinth, string courses, foundations, footings, bases of columns, suspended floors, roofs, landings, balconies and access platforms, etc.

xv. Providing solid masonry using pre-cast solid blocks (factory made) of size 40x20x20cm or nearest available size confirming to IS 2185 part I of 1979 for super structure up to floor two level with thickness 20cm in: CM 1:6 (1 cement: 6 coarse sand) etc. complete.

xvi. Providing steel reinforcement for R.C.C work including straightening, cutting, bending, placing in position and binding all complete using Thermo - Mechanically Treated bars of grade Fe-500D or more.

xvii. Supplying and fixing rolling shutters of approved make, made of required size M.S. laths, interlocked together through their entire length and jointed together at the end by end locks, mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete, including the cost of providing and fixing necessary 27.5 cm long wire springs manufactured from high tensile steel wire of adequate strength conforming to IS: 4454 - part 1 and M.S. top cover of required thickness for rolling shutters.80x1.25 mm M.S. laths with 1.25 mm thick top cover.

xviii. Providing steel work welded in built up sections/framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required in gratings, frames, guard bar, ladder, railings, brackets, gates and similar works.

xix. Supplying and fixing Rectangular C.I. manhole covers 455mm x 610mm with frame (low duty).

xx. Supplying and Providing MS dowel bars of size 25 mm dia of 2.0m long (1m in rock and 1m in concrete) including drilling holes of 30mm dia and filling the gap with cement grout(0.50kg/each) etc.

xxi. Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS : 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/ neoprene gasket

etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminnium snap beading for glazing /paneling, C.P. brass/ stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge.

xxii. Providing and fixing M.S. Grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer.

xxiii. Construction of Service Reservoir, including related earthwork, dismantling any structure, construction of suitable type of foundations suitable to the soil conditions and loading sustained, providing and fixing inlet, outlet, overflow and scour pipes, vent pipes, valve chambers, overflow and scour chambers, anchor blocks, scour pit inside the tank, clear water drain from treatment plant to new sump including earth work, RR support block and concrete channel with cover slab length 30 m and size of 60x60 cm etc. as per design and construction drawings showing detailed reinforcement details and sizes which shall be submitted by contractor after getting proof checking of design. The design shall be got approved by the Tendering authority before construction work is started.

xxiv. The scope also includes providing and fixing inlet valves, outlet valves, overflow valves, scour valves with PN1.6 rating having same size as that of the pipe with Valve Chambers are included in the scope of the work and the rate quoted shall be inclusive of the same.

xxv. Providing and fixing 75 mm or nearest size CI vent cowl and specials covered with steel net as per approved design and drawings. A minimum of 1 no of vent cowl shall be provided.

xxvi. Finishing walls with Premium Acrylic Smooth exterior paint with Silicone additives of required shade:New work (Two or more coats applied @ 1.43 ltr/ 10 sqm over and including priming coat of exterior primer applied @ 2.20 kg/ 10 sqm).

xxvii. The scope also includes painting the name of the scheme, logo and other details on the reservoir as per the direction of the Engineer in charge.

xxviii. Supplying and fixing of level indicator and other necessary fixtures as required and directed by Engineer In charge.

xxix. Testing of the Reservoirs for water tightness by filling the tank using portable water.

xxx. The scope of work also includes cleaning, washing and disinfection of inside of reservoirs making it free from any foreign matter, loose particles, debris, etc. and making it suitable for storage of potable water, commissioning and handing over the

reservoir and piping works with all other items.

xxxi. Submission of as built drawings, Reports, designs and drawings etc. connected with the work is also included in the scope. These shall be furnished by the contractor for approval, minimum in triplicate copies or as called for by the Engineer in charge. Contractor shall prepare and submit to the approving authority six additional copies of the approved drawings immediately on receipt of approval for transmission to various field offices. On completion of approval of all items connected with the project, the contractor shall prepare three copies of bound volumes of the approved documents and furnish to the approving authority. The preparation and binding of this document should be in such a manner that during the long period of future reference and recorded life, it shall not get disintegrated nor legibility is lost. The layout and Design of all components of GLSR shall be prepared and got approved from the agreement authority before execution of the work without any extra cost.

2.1.2. Supplying, laying and Testing of Clear Water Pumping Main

The scope of this part of the work covers procurement of following pipes with rubber rings and the required specials & valves conforming to the required standards and laying them along the proposed alignments. All the works connected with supplying, laying and testing of Clear Water Pumping Mains have to be included by the tenderers in their bids.

Sl No	Size	Class of Pipe	
1.	150mm DI K9	DI	K9(Tyton)
2.	200mm DI K9	DI	K9(Tyton)
3.	250mm DI K9	DI	K9(Tyton)

Valves			
Sl No	**Size**	**Class of Pipe**	**Quantity**
1.	40mm Air valve single orifice	Type S1	5
2.	50mm Air valve double orifice	Type DS2	17

3.	80mm CI Air valve	Kinetic Air Valve Type DS1	24
4.	150mm Double Flanged Sluice valve with Cap	PN 1.6	2
5.	200mm Double Flanged Sluice valve with Cap	PN 1.6	2
6.	250mm Double Flanged Sluice valve with Cap	PN 1.6	4

DI Specials

7.	DI Special (as per site condition)		65.960 quintal		
MJ Collar					
8.	150 mm DI		17.670 quintal		
9.	200 mm DI				
10.	250 mm DI				

Valve chamber

Sl No	**Size**	**specification**	**Quantity (No)**
1.	Valve chamber	1.3m x 1.3 m x 1.3m	6
2.	Valve chamber	1m x 1m x 1.3m	2

2.1.3. The work abstracted in clause 2.1 above includes the completion of works specified below

1. Supplying and laying 150mm, 200mm and 250mm dia DI K9 pipes using with rubber rings, tees, bends and specials, trenching in all classes of soil, jointing according to IS specification, testing with required hydraulic pressure, back filling the trenches, connecting & commissioning including providing fencing, shoring, lighting, watching, providing caution board, dewatering, anchor blocks, red earth cushioning wherever necessary including cutting asphalt/concrete surface road etc. wherever necessary as per direction of departmental officers.

2. Supplying and fitting 150mm, 200mm and 250mm CI D/F (PN 1.0) Sluice valves with cap conforming to relevant BIS/ISO standards and bearing ISI /ISO marking for control valves/scour valves with required specials as per the

conditions of contract and specifications. List of Valves are given as Annexure.

3. Supplying and fitting 40mm single small orifice type S1, 50mm CI air valve double orifice type DS2 conforming to IS 14848 - 2000 and 50mm CI air valve with flanges, Kinetic air valve type DS1 conforming to IS 14845 - 2000 fitted with bolts, nuts, rubber insertions etc.

4. Construction of R.C.C valve chambers cost of excavation work, 100 mm thick concrete base, 250 mm side wall, 150 mm bottom and 250 mm cover slab of chamber concrete with reinforcement etc. to be completed as per approved drawing for air valves & sluice and scour valves of size 100mm and above, including earth work excavation in all classes soil as per the direction of departmental officers

5. Cutting the bituminous / concrete roads with cutting machine for a minimum depth of 200mm along the sides of proposed alignment of the pipe to be laid without causing any damage to other utilities, including lighting, ribbon fencing, caution boards, traffic diversion, and as per the direction of departmental officers etc. complete, before carrying out the demolition of bituminous / concrete road by mechanical means and carrying out the excavation.

6. Dismantling manually / by mechanical means and disposal of material within 50 metres lead as per direction of Engineer -in-Charge (for Bituminous road).

7. Demolishing cement concrete manually / by mechanical means including disposal of material within 50 metres lead as per direction of Engineer - in-Charge. Nominal concrete 1:3:6 or richer mix (i/c equivalent design mix)

8. Fabricating suitable size M.S. pipes and flanges for covering the proposed pipes for crossing culvert/ bridge portions. This also include cutting, welding and grinding in M.S. pipes.

9. Design and construction of R.C.C/ C.C anchor block including earth work excavation in all classes of soil wherever required, as per the direction of departmental officers and approved drawing

10. Protection works to electric posts, existing line/cable/pipe line/any underground utilities etc wherever required

11. Providing sufficient spare pipes for the future maintenance of the pipeline shall be supplied as spares as given in the price schedule.

2.1.4 Road Restoration

1. Maintenance of Earthen Shoulder (filling with fresh soil) Making up the loss of material/ irregularities on the shoulder to the design level by adding fresh approved soil and compacting it with appropriate equipment.

2. Removal of unserviceable soil including excavation, loading and disposal upto 1000 metres lead but excluding replacement by suitable soil which shall be paid separately as per clause 305.

3. Excavation for road work in soil with a hydraulic excavator of 0.9 cum bucket capacity including cutting and loading in tippers, trimming bottom and side slopes, in accordance with requirements of lines, grades and cross sections and transporting to embankment location within all lifts and lead up to 1000m as directed by the Engineer in Charge.

4. Construction of granular sub-base by providing graded material, spreading in uniform layers with a motor grader on a prepared surface, mixing by mix in-place method with rotavator at OMC, and compacting with a vibratory roller to achieve the desired density, complete as per clause 401. Grading-III -For lower sub-base - Mix in Place Method.

5. Providing, laying, spreading and compacting graded stone aggregate to Wet Mix Macadam specification including premixing the Material with water at OMC in mechanical mix plant carriage of mixed Material by tipper to site, laying in uniform layers with paver in sub- base / base course on well prepared surface and compacting with vibratory roller to achieve the desired density

6. Providing and applying primer coat with bitumen emulsion (SS) on prepared surface of granular Base including clearing of road surface and spraying primer at the rate of 0.70 - 1.0 kg/sqm using mechanical means

7. Providing and applying tack coat with bitumen emulsion (RS) using emulsion pressure distributor at the rate of 0.25 - 0.30 kg per sqm on the prepared Granular Surface cleaned with mechanical broom

8. Providing and laying bituminous macadam with 80-100 TPH hot mix plant producing an average output of 75 tons per hour using crushed aggregates of specified grading premixed with a bituminous binder (VG 30), transported to the site, laid over a previously prepared surface with paver finisher to the required grade, level, and

alignment and rolled as per clauses 501.6 and 501.7 to achieve the desired compaction for Grading II - (19 mm nominal size)

9. Providing and applying tack coat with bitumen emulsion (RS) using emulsion pressure distributor at the rate of 0.20 - 0.30 kg per sqm on the prepared bituminous surface cleaned with mechanical broom

10. Providing and laying bituminous concrete with 80-100 TPH hot mix plant producing an average output of 75 tons per hour using crushed aggregates of specified grading, premixed with a bituminous binder(NRMB) @ 5.2 percent of mix and filler, transporting the hot mix to work site, laying with a hydrostatic paver finisher with sensor control to the required grade, level, and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MORTH specification clause No. 507 complete in all respects For Grading - I (19 mm nominal size)

11. Construction of granular sub-base by providing graded material, spreading in uniform layers with a motor grader on a prepared surface, mixing by mix in-place method with rotavator at OMC, and compacting with a vibratory roller to achieve the desired density, complete as per clause 401. Grading-IV - For lower sub-base - Mix in Place Method

12. Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40 mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days

13. Plain/Reinforced Cement Concrete in Open Foundation complete as per Drawing and Technical Specifications PCC Grade M20.

2.1.5 Road Restoration

i) Restoration of earthen berm.

a) Earth work excavation

The work aims to restore the damage caused during earthwork excavation for pipe laying for a depth upto 1.5m. The work includes removal of unserviceable soil for a depth of 0.2m including excavation, loading and disposal upto 1000 metres lead but excluding replacement by suitable soil which shall be paid separately as per clause 305 and maintenance of earthen

shoulder (filling with fresh soil) making up the loss of material/ irregularities on the shoulder to the design level by adding fresh approved soil and compacting it with appropriate equipment.

b) Providing Sub base with GSB

After carrying out necessary earth work excavation, the trenches have to be filled with granular sub base for a depth of 0.2m after necessary compaction. Construction of granular sub-base by providing close graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per clause 401.

ii) Restoration of concrete pavement

a) Earth work excavation

The work includes earth work excavation removal of loose soil from the trenches to a depth of 0.4m for PWD roads and 0.6m for SH concrete shoulder from the road surface level.

b) Providing Sub base with GSB

After carrying out necessary earth work excavation, the trenches have to be filled with granular sub base for a depth of 0.25m after necessary compaction.

c) Providing Plain Cement Concrete

This work involves Plain cement concrete 1:3:6 nominal mix in foundation with crushed stone aggregate 40 mm nominal size mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days for a depth of 0.15m. This also includes Plain/Reinforced Cement Concrete in Open Foundation complete as per Drawing and Technical Specifications PCC Grade M20 (Without formwork) for a depth of 0.075m. This includes curing of the concrete surfaces as per relevant IS standards.

iii) Restoration of bituminous surface

2.3.4.1. Restoration of NH and SH

a) Earth work excavation

The work includes earth work excavation removal of loose soil from the trenches to a depth

of 0.2m from the road surface level.

b) Providing Sub base with GSB

After carrying out necessary earth work excavation, the trenches have to be filled with granular sub base for a depth of 0.25m after necessary compaction.

c) Providing base with WMM

After carrying out necessary earth work excavation, the trenches have to be filled with wet mix macadam for a depth of 0.25m after necessary compaction.

d) Providing prime coat

Providing and applying primer coat with bitumen emulsion (SS) on prepared surface of granular Base including clearing of road surface and spraying primer at the rate of 0.70 - 1.0 kg/sqm using mechanical means.

e) Providing tack coat

Providing and applying tack coat with bitumen emulsion using emulsion pressure distributor at the rate of 0.20-0.30 kg per sqm on the prepared bituminous and granular surface cleaned with mechanical broom.

f) Providing bituminous macadam

Providing and laying bituminous macadam for a thickness of 0.05m, with 80-100 TPH hot mix plant producing an average output of 75 tonnes per hour using crushed aggregates of specified grading premixed with bituminous binder (VG 30), transported to site, laid over a previously prepared surface with paver finisher to the required grade, level and alignment and rolled as per clauses 501.6 and 501.7 to achieve the desired compaction. For Grading II - (19 mm nominal size)

g) Providing bituminous concrete

Providing and laying bituminous concrete for a thickness of 0.03m with 80-100 TPH batch type hot mix plant producing an average output of 75 tonnes per hour using crushed aggregates of specified grading, premixed with bituminous binder @ 5.4 per cent of mix and filler, transporting the hot mix to work site, laying with a hydrostatic paver finisher with sensor control to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MORTH specification clause No. 507 complete in all respects.

Restoration of Major District Roads (PWD Roads):

a) Earth work excavation

The work includes earth work excavation removal of loose soil from the trenches to a depth of 0.7m from the road surface level.

b) Providing Sub base with GSB

After carrying out necessary earth work excavation, the trenches have to be filled with granular sub base for a depth of 0.25m after necessary compaction.

c) Providing base with WMM

After carrying out necessary earth work excavation, the trenches have to be filled with wet mix macadam for a depth of 0.25m after necessary compaction.

d) Providing prime coat

Providing and applying primer coat with bitumen emulsion (SS) on prepared surface of granular Base including clearing of road surface and spraying primer at the rate of 0.70 - 1.0 kg/sqm using mechanical means.

e) Providing tack coat

Providing and applying tack coat with bitumen emulsion using emulsion pressure distributor at the rate of 0.25-0.30 kg per sqm on the prepared granular surface cleaned with mechanical broom

f) Providing close graded premix surfacing material

Providing, laying and rolling of close-graded premix surfacing material of 20 mm thickness composed of 11.2 mm to 0.09 mm (Type-A) aggregates using viscosity grade bitumen (VG - 30) to the required line, grade, and level to serve as wearing course on a previously prepared base, including mixing in a suitable HMP of appropriate capacity not less than 75 tonnes/hour., laying and rolling with a Smooth wheeled roller 8-10 tonne capacity, and finishing to the required level and grade.

2.1.6 Site Condition

There are few narrow reaches where road may have to be protected during excavation and compaction. Care should be taken for preventing any kind of damage to the existing portion

of the road or any other utilities. Care should be taken to avoid damages to utilities of KWA. The site conditions are generally good and no other major hurdles are anticipated.

2.1.7 Storage of materials, plant and tools

The conveyance, loading, unloading and stacking of materials for the project shall be the responsibility of the contractor. Materials supplied for the work shall be stacked at the site with maximum care to avoid any damages to the pipe and its protective coatings. The materials shall not be stacked on roadside in such a way as to cause hindrance to normal traffic. If there is storage space available at KWA yards, the contractor shall stack the materials at the site as per direction of the Executive Engineer. The balance materials, after completion of works shall be brought to KWA Division store as per direction of the Engineer in charge.

2.2. STANDARD SPECIFICATIONS

2.2.1 Laying Pipes and Specials

1. Earthwork Excavation

i. Excavation for the pipe line work is to be done generally in hard soil, medium rock, or ordinary rock. Hard rock is also present along the proposed alignment of the pipe line where blasting is prohibited. Wherever blasting is done for trenching, protection shall be provided with sand bag to void damages to nearby structures. Blasted rock recovered during excavation shall be stacked and handed over to the department.

ii. A 20 cm thick layer of red earth cushion is to be provided in the pipes trenches where the excavated bed of trench is rocky. Excavation shall also be carried out for cutting of the existing pavement formation, where the asphalt surface may have to be cut and removed. Care shall be taken to ensure that damages to the road surface are the least and hindrance to vehicular and pedestrian traffic is minimum. There may be other utilities lying along the proposed pipe line alignment such as telephone cables, electricity cables, water supply lines etc; and contractor shall have to take maximum care in excavation for laying of the pipe line. If any of the existing structure/utilities are damaged during the excavation for pipe laying works, the contractor shall have to make good the losses/pay the claim of the respective department and Kerala Water Authority will not be responsible for any such expenses incurred by the contractor.

iii. Excavation for laying of pipes shall have to be done in lines so as to provide a minimum cover of 1m above the pipes to be laid and care shall be taken to avoid sudden change in gradient and alignment. The requirement of minimum cover may increase up to 1.20m where the pipes are to be laid over PWD/main highway roads. Also the depth of excavation may increase where the pipe line shall have to cross existing utilities or culvers or drains as decided by the Engineer in charge, based on specific site conditions. Sufficient width as per standard specification and KWA Technical Circular shall be provided. Socket trench shall be taken at every joint before laying pipes so as to enable the body (shell) of the pipe to rest on leveled compacted firm ground in the trench

iv. Only select fill materials free of stones and sharp aggregates shall be used for refilling the trenches to a height of 30cm from the top of pipeline. Necessary earth shall be supplied by the tenderer without any extra cost if the excavated earth is found insufficient for filling the trenches. All the backfilling shall be watered and compacted in 20cm layers to avoid further settlement of refilled portion. Any excess of soil after refilling the trenches shall be removed by the contractor and disposed off from site within a distance of 5km as directed by the Executive Engineer. If the site condition so warranted, the excavated earth shall be conveyed to a convenient place as directed by the engineer to a dumping place within a distance of 5 KM for avoiding inconvenience to normal traffic and the same shall be transported back for refilling without any extra cost.

v. Protection shall be provided for the filled up earth wherever necessary and as directed by the Engineer in Charge.

vi. In case the contractor fail to provide necessary protection work such as fencing, shoring etc. as required at the site, the Engineer in charge will have the power to deduct the charges of such work at department rates from the payments due to the contractor for the work done by him at such sections. Contractor shall arrange for necessary bailing out water for trenching and pipe laying works and the rate quoted by the tenderer shall be inclusive of all incidental charges such as shoring and any other protection to the sides of trenches. Protections to the telephone and electric post, convenient re- alignment of existing utilities within the proposed alignments, taking all site safety measures such as providing fencing, lighting and watching of the worksite, bailing out water etc. complete should be done by the contractor. No claim for any extra items will be entertained against any of the incidental changes that may arise out of contingencies associated with the completion of the work

vii. The contractor shall take care to minimize the damages to road surface while trenching is done for pipes laying and construction of valve chambers, anchor blocks etc. Damages to structures like culverts, retaining walls, buildings etc. caused during pipe laying works shall be fully rectified by the contractor at his own cost to the satisfaction of the respective owners of the structures. Compensation if any for such damages shall be met by the contractor alone. Contractor shall use asphalt cutters to minimize damages to tar surfaces. Road crossing portions, the backfilling

shall be watered and compacted in 20cm layers to avoid further settlement of refilled portion and shall be provided with concrete 1:2:4 for a thickness of 25 cm in level with the existing road surface.

2. Laying pipes & specials

The pipe shall be laid in accordance with approved layout plan. However the Executive Engineer has the authority to change the alignment for better benefits to public and department during execution. In such cases the contractor shall lay pipelines as per direction of the Executive Engineer without any additional cost.

Pipes shall be laid in straight line in trenches so that the body directly rests on the soil bed. Handling of pipes to lay in trenches or in any support structure must be done carefully and proper equipments should be used for the purpose. Any deviation from straight line shall be done using standard bends. In case where standard bends are not available for any particular shape of bend, specials fabricated with MS sheets of thickness not less than 8 mm shall be used with the prior approval of the Engineer in charge. All culverts along the alignment of the pipe line are to be crossed with GI pipes and the bridge crossing shall be with GI / M.S.pipe, as per the instructions of Engineer in charge and as per approved standards and specifications.

Pipes shall be laid in trenches as per the standard specifications described in respective IS codes for laying pipes. The contractor shall follow the specifications of the IS codes with latest amendments regarding laying of pipes.

3. Jointing

Jointing of the pipes shall be as per the IS specification and should be as per the direction of the departmental officers. Jointing of the DI K9 pipe shall be with Tyton rubber rings/lead joint. The rubber ring shall be of best quality and approved make and shall confirm to the relevant IS Code. The rubber ring shall not be prone to termite or any other such insect attack. Jointing of valves shall be in such a way that the removal is easy. Use of mechanical joints of reputed make will be appreciated

4. Road Crossings and Bends

All road crossing shall have to be planned in advance and carried out with the concurrence of the Engineer in charge. As far as possible 125 mm to 80 mm, 90degree bends should be avoided in the road crossings and road bends etc. Contractors will not be permitted to take angular deviation at pipe joints more than 2/3rd

of the maximum angle, provided as “allowable” in the relevant specification / product information.

All necessary sanctions for laying pipes along NH/SH through fares and road crossings shall be obtained by the Department and the roads and other structures are to be made back to their original condition after laying of pipes except re-tarring of road surface. The road crossing of Panchayath roads should be filled red earth and consolidated in layers after watering and ramming, etc. Then the trench should be filled with cement concrete 1: 2: 4 for a minimum depth of 25cm so that the top surface is in level with existing road surface. The same shall be applicable to the edges of roads where severe damages occurred during trenching.

5. Crossing of bridges and culverts

Crossing of bridges and culverts shall be done by DI flanged pipes/GI pipes/M.S. pipes as the case may be and should be laid exposed. Where the pipe line crosses over the culvert an air valve shall be provided at the crust to avoid air block. Care shall be taken to avoid 90º bends. Culvert crossings may also have to be done at reaches where the above methods are not possible by carrying the pipe line outside the culvert body and supporting it on RCC pillars but without obstructing the vent way. Protection to the exposed pipes in the form of anchor blocks or casing pipes shall be provided.

The contractor shall consult with the local NH/PWD/local body authorities to finalize the alignment and method of pipe laying at culvert and bridge crossings. The discretion in this regard shall however-be finalized by the Engineer in charge. The contractor shall assess the actual number of culverts/bridge at site before supplying and laying pipes

6. Fixing valves

Control valves, scour valves and air valves shall be fitted in pipe lines at appropriate locations as suggested by the Engineer in charge. All scour lines are to be laid with scour discharge line of required size flanged CI Class A/DI pipe with sufficient lead length not less than 2.00 m.

7. Valve Chambers

RCC valve chambers shall be constructed for all valves of size 80 mm & above, except for air valves provided on exposed pipes at culvert & bridge crossings. Valve chambers shall be provided with a foundation 20 cm thick (in CC 1:3:6 using 50%

40mm broken stone & 50% 20mm broken stone) and RCC cover slab, thickness be not less than 20 cm (in RCC 1:1.5:3 using 20mm broken stone) capable to take load of vehicular traffic designed for class AA loading. For valves of size up to 125mm, valve chambers of inside dimension 120 cm x 120 cm with wall thickness be not less than 20 cm (in RCC 1:2:4 using 20mm broken stone) shall be provided. For valves of size above 150 mm, the inside dimension shall be 150 cm x 150 cm and wall thickness shall be not less than 20 cm (in RCC 1:2:4 using 20mm broken stone). The height of the valve chamber shall be such that the top of cover slab is in level with the road surface. All the valve chambers shall be neatly plastered inside with CM 1:3 and there shall not be any seepage from bottom or sides. Valve chambers deeper than

1.20 m shall be provided with Cl foot rest on sides. The design of valve chambers shall be submitted to the Executive Engineer and got approved before commencing construction. Foundation to support the valves in CC 1:1.5:3 shall be provided. The contractor should assess the actual no of valve chambers required as per site inspection.

8. Anchor blocks & supports

Anchor blocks in CC 1:2:4 as per the standards and specification of KWA/CPHEEO shall be provided in pipe lines at bends, steep slopes and at places as directed by the Engineers in charge. Typical designs shall be got approved by the Engineer in charge before the actual construction. Clamps, wherever necessary, shall be provided with MS flats of minimum 4 mm thickness and 40 mm width and painted with epoxy paints. When pipe lines are to be laid outside of a culvert or bridge, as insisted by the PWD officials, proper support pillars in RCC 1:1.5:3 shall be provided. The design of such support pillars shall be got approved by the Executive Engineer before the construction.

9. Hydrostatic pressure testing of pipe lines at field.

After laying the pipe lines, hydrostatic pressure testing shall be done in reaches of not more than 200 m in length as per the relevant IS specifications to ensure that the pipes and joints are sound enough to withstand the maximum pressure likely to be developed under working conditions. Contractor shall not be allowed to proceed with the pipe laying work leaving more than 400m (i.e. 2 reaches) of pipelines untested. The joints of all pipes and specials shall be kept open until the testing is completed satisfactorily. The water required for testing the pipe lines, dummy pipes and test pumps etc. shall be arranged by the contractor with his cost. Hydrostatic pressure testing on pipe lines shall be conducted as per the norms of CPHEEO Manual

on Water Supply and Treatment (1999). Failure of test due to any reason is to be rectified and retested without any extra cost. Contractor shall maintain a record of testing of pipeline which shall be got signed by an officer not below the rank of an Assistant Executive Engineer which shall be produced along with his request for release of security deposit and shall be presented to any inspecting officer above the rank of Assistant Executive Engineer. Positive results observed during pre delivery inspection and the acceptance of the inspection report / result by Kerala Water Authority/Agreement authority will not anyway relieve the contractor of his responsibility on the quality of materials supplied by him at the work site. If the materials supplied at site are found to be of inferior quality, the Executive Engineer in charge will have all the right to ask the tenderer to replace the materials supplied at his cost.

10. Fabrication and welding

Any fabrication work for bends and specials shall be with minimum 8 mm thick MS plates. The drawing pertaining to such works shall be got approved by the Engineer in charge before the actual fabrication. The schedule of such shop/site of fabrication shall be intimated to the Engineer in charge so as to enable him to inspect the fabrication and welding works. The contractor shall submit welding procedure to the Engineer along with the data regarding certification, experience and other relevant details of the welders and got approved by the Engineer, before the commencement of any welding work for steel structures or specials. The corrosion protection planned for the fabricated components shall also be reported to the Engineer in charge and got approved. The location of the welding shop would be within the area of work and shall be easily accessible for the inspecting officers

11. Supply of specials

The tenderer may also supply pipe manufactured as per I.S.O. standard but in such case the guarantee period shall be 60 months. The DI pipe shall be shop lined, centrifugally with cement mortar inside and coated with metallic zinc with bituminous finishing layer outside, all confirming to IS standards. The M.S. Pipes shall be provided with epoxy coating. The work includes supplying and fixing all D.I and M.S specials required for the work including bends, tees, etc, confirming to relevant IS Codes. The specials shall also be lined with metallic zinc with bituminous finishing layer outside confirming to relevant IS Codes. The M.S. specials shall have epoxy coating. The cost of pipe shall be quoted inclusive of the cost of all specials and jointing materials such

as labour charges, hemp, white lead etc. Necessary flanged bends according to the site conditions are to be provided. Care shall be taken to avoid bends 90º in the line. DI specials to suite the respective DI (K9) pipes such as bends, Tees, Mechanical joints, Tapers, Flanged sockets etc shall be supplied as per actual requirements at site after getting approval of Executive Engineer.

12. Vendor List

Two copies of complete list of Vendors/Manufacturers from whom the valves, specials etc. can be procured with the product specification, Brochure/ Product identification number etc. shall be submitted by the contractor on completion of the work.

Note: - The above points will give only general information of the work to be executed under this contract. The bidders shall meticulously inspect the proposed alignment, survey the required areas, water courses which the pipeline will have to cross, road crossing etc. and assess the items of materials to be supplied and works to be done, for satisfactory completion.

3. PRE–DELIVERY INSPECTION OF PIPES AND OTHER MATERIALS

The contractor shall supply all the materials including pipes and specials covered under the contract at his own cost. The pipes, specials etc. should be tested at the factory by the third party inspection agencies authorized by Kerala Water Authority in presence of a technical officer of Kerala Water Authority deputed by the agreement authority, for witnessing the pre delivery testing by the third party inspectors. In the case an officer from Kerala water Authority deputed for witnessing the pre delivery inspection of materials at factory, the TA &DA of the officer will be paid by the Authority, if the test results are acceptable and positive. In case if the test results are negative, all such expenses of deputing the Kerala Water Authority officer of the pre delivery inspection shall have to be met by the contractor. Inspection certificate should be produced before the dispatch of materials from the factory and only after written communication/after the verification and acceptance of the inspection report, the materials shall be dispatched to the site. The third party inspection is mandatory for all pipes, specials, valves and all electro mechanical equipments supplied by the contractor. Also ISI marked materials shall alone be accepted, wherever ISI specifications is available for such items. Any dispute between the contractor and the supplier or the inspection agencies shall be a matter of settlement among them and the Kerala Water Authority shall not be a party to such dispute. The lump sum amount quoted shall be inclusive of the cost of all materials and charge of

inspection as specified herein

4. SAFETY AT SITE

Tenderers alone shall be responsible for any damage to men and materials and that may occur at site due to lapses in safety measures. Tenderers shall provide necessary caution boards and fencing as per the directions/specifications of KWA for trenches during the progress of work. Minimum 3 Nos. caution boards shall be placed in the approaching side and 2 Nos. at the leaving side at the required distance as per the relevant safety codes. They shall also provide sufficient shoring to the sides of trenches to prevent collapse of the sides of the trenches and consequent damages to the men and materials whenever the depth of trench exceeds 1.50m or while excavating near other civil structures. During night hours, contractors shall provide sufficient lighting and watching at the site. Failure to adhere to these clauses shall attract fine and penalties by way of deductions from the payments to contractors at an amount worked out at department rates

5. REPORTS, DESIGNS AND DRAWINGS

1. Reports, designs and drawings etc. connected with the work shall be furnished by the contractor for approval, minimum in triplicate copies or as called for by the Engineer in charge.

2. Contractor shall prepare and submit to the approving authority seven additional copies of the approved drawings immediately on receipt of approval for transmission to various field offices.

3. On completion of approval of all items connected with the project, the contractor shall prepare three copies of bound volumes of the approved documents and furnish to the approving authority. The preparation and binding of this document should be in such a manner that during the long period of future reference and recorded life, it shall not get disintegrated nor legibility lost.

6. AS LAID MAPS

Contractor shall submit five hard copies and soft copies in CD, of the As laid map of the entire facility he had installed for the project as part of this contract with necessary references to the existing permanent structures. The As laid map must be prepared progressively with the progress of work and must be presented along with all the running contract bills. Submission of the As laid map in complete- for the entire work is mandatory for considering the work as completed.

7. MATERIALS FOR STRUCTURES

7.1. SCOPE

1. Materials to be used in the work shall conform to the Specifications mentioned on the Drawings, the requirements laid down in this section and Specifications for relevant items of work covered under these Specifications.

2. If any material, not covered in these specifications, is required to be used in the work, it shall conform to relevant Indian Standards, if there are any, or to the requirements specified by the Engineer.

7.2. CEMENT

1. Cement to be used in the works shall be any of the following types with the prior approval of the Engineer:

a. Ordinary Portland cement, 33 Grade, conforming to IS: 269-1989 or latest revision.

b. Rapid Hardening Portland Cement, conforming to IS: 8041-1990 or latest revision

c. Ordinary Portland cement, 43 Grade, conforming to IS: 8112-1989 or latest revision.

d. Ordinary Portland cement, 53 Grade, conforming to IS: 12269-1987 or latest revision.

e. Sulphate Resistant Portland cement, conforming to IS: 12330-1988 or latest revision

Cement conforming to IS: 269-1989 or latest revision shall be used only after ensuring that the minimum required design strength could be achieved without exceeding the maximum permissible cement content of 450kg/cum. of concrete

7.3. REINFORCEMENT STEEL

Reinforcement steel bars HYSD / TMT bars of grade Fe-500D or more conforming to IS: 1786 shall be used. The tenderer shall inform the Engineer in charge prior to start of work, regarding the type of steel to be used by him, in the works and provide the samples of all diameter of bars used at site. Necessary test as directed by Engineer in charge should be carried out at reputed institution as directed by the engineer in charge in the samples to affirm the quality. The cost of all such testing shall be borne by the contractor. Only the bars approved by him shall be used in the work,

7.4. COARSE AGGREGATES

For plain and reinforced cement concrete (PCC and R C C) o r prestressed concrete (PSC) works, coarse aggregate shall consist of clean, hard, strong, dense, non-porous and durable pieces of crushed stone, crushed gravel, natural gravel or a suitable combination thereof or other approved inert material. They shall not consist pieces of disintegrated stones, soft, flaky, elongated particles, salt, alkali, vegetable matter or other deleterious materials in such quantities as to reduce the strength and durability of the concrete, or to attack the steel reinforcement. Coarse aggregate having positive alkali silica reaction shall not be used. All coarse aggregates shall conform to IS: 383-1970 and tests for conformity shall be carried out as per IS: 2386-1963 Parts I to VIII or latest revision

7.5. SAND/FINE AGGREGATES

For masonry work, sand shall conform to the requirements of IS: 2116-1980 or latest revision. For plain and reinforced cement concrete (PCC and RCC) or prestressed concrete (PSC) works, fine aggregate shall consist of clean, hard, strong and durable pieces of crushed stone, crushed gravel, or a suitable combination of natural sand, crushed stone or gravel. They shall not contain dust, lumps, soft or flaky, materials, mica or other deleterious materials in such quantities as to reduce the strength and durability of the concrete, or to attack the embedded steel. Motorized sand washing machines should be used to remove impurities from sand. Fine aggregate having positive alkali-silica reaction shall not be used. All fine aggregate shall conform to IS: 383-1970 and test for conformity shall be carried out as per IS: 2386-1963 (Part I to VIII) or its latest revision. The Contractor shall submit to the Engineer the entire information indicated in Appendix A of IS: 383-1970 or latest revision. The fineness modules of fine aggregate shall neither be less than 2.0 nor greater than 3.5.

7.6. WATER

Water used for mixing and curing shall be clean and free from injurious amounts of oils, acids, alkalis, salts, sugar, organic materials or other substances that may be deleterious to concrete or steel. Potable water is generally considered satisfactory for mixing concrete. Mixing and curing with sea water shall not be permitted and pH value shall not be less than 6.

7.7. TESTS AND STANDARD OF ACCEPTANCE

All materials, even though stored in an approved manner shall be subjected to an acceptance test prior to their immediate use. Independent testing or cement for every

consignment shall be done by the Contractor at site in the laboratory approved by the Engineer before use. Any cement with lower quality than those shown in manufacturer's certificate shall be debarred from use. In case of imported cement, the same series of tests shall be carried out before acceptance.

7.8. TESTING AND APPROVAL OF MATERIAL

The Contractor shall furnish test certificates from the manufacturer/supplier of materials with each batch of material(s) delivered to site.

The Contractor shall set up a field laboratory with necessary equipment for testing of all materials, finished products used in the construction as per requirements of conditions of contract and the relevant Specifications. The testing of all the materials shall be carried out by the Engineer or his representative for which the Contractor shall make all the necessary arrangements and bear the entire cost.

Tests which cannot be carried out in the field laboratory have to be got done at the Contractor's cost at any recognized laboratory/testing establishments approved by the Engineer

7.9. SAMPLING OF MATERIALS

Samples provided to the Engineer or his representative for their retention is to be in labeled boxes suitable for storage.

Samples required for approval and testing must be supplied well in advance by at least 48 hours or minimum period required for carrying out relevant tests to allow for testing and approval. Delay to works arising from the late submission of samples will not be acceptable as a reason for delay in the completion of the works.

If materials are brought from abroad, the cost of sampling/testing whether in India or abroad shall be borne by the Contractor.

7.10. REJECTION OF MATERIALS NOT CONFORMING TO THE SPECIFICATIONS

Any stack or batch of material(s) of which sample(s) does not conform to the prescribed tests and quality shall be rejected by the Engineer or his representative and such materials shall be removed from site by the Contractor at his own cost. Such rejected materials shall not be made acceptable by any modifications.

7.11. TESTING AND APPROVAL OF PLANT AND EQUIPMENT

All Plants and equipment used for preparing, testing and production of materials for incorporation into the permanent works shall be in accordance with manufacturer's

Specifications and shall be got approved by the Engineer before use.

7.12. HANDLING OF BALANCE MATERIALS

Balance materials if any should be conveyed by the contractor in his own cost to the respective KWA, Division store before closing the final bill.

LIST OF CODE OF PRATICE FOR REFERENCE

Sl. No.	Description	Standards
1	HDPE Pipe	IS: 4984/1995
2	DI Pipes	IS 8329/2000
3	PVC Pipe	IS 4985/1988, IS 4985/2000
4	G I Pipes	IS 1239
5	Sluice Valve	IS 14846/2000 & IS 2685/1971
6	Air Valve	IS 4845/2000
7	Cement	IS 269/1976 & IS 8112/1989
8	Steel Bars	IS 432-Part I, IS 1139 & IS 1786
9	Sand	IS 2116/1985 & IS 1542/1977
10	Water for mixing & Curing	4.3 of IS 456/2000
11	Coarse & fine aggregates for concrete	IS 383/1970 & IS 456/2000
12	Admixture	IS 9103/1972
13	DI & PVC Specials	IS 9523/2000 & IS 7834/1987
14	Non standard MS bend	IS 816/1969 & IS 822/1970 & IS 823/1964
15	Rubber Rings	IS 12820/2004 & IS 5382/1985
16	Solvent cement for joints	IS 7634/2003 – Part III
17	Anti corrosive epoxy painting for	IS 6916/1973
18	Pipe line pressure test	IS 12288/1987
19	Lightning arrestors	IS 4850
20	Drinking water standards	IS 10500/1983
21	Vaccum type chlorinators	IS 10553/1983 – Part I & II
22	Steel structures	IS 800
23	Sluice Valve with gear	IS 780/1984
24	Testing of concrete	IS 1199/1959 & IS 516/1959 & IS 456/2000
25	Mixing of concrete	Clause 8 & 9 of IS 456/2000
26	Code of practice for concrete structures for	IS 3370/2009 – Part I to IV
27	Centrifugal pump	IS 1520/1980
28	Motors	IS 325/1978 & IS 900/1965 & IS 4029/1967
29	Trenching	PHED data Item No. 1001, 1002 & 1008
30	Non metallic gasket	ASA B 16.21
31	Criteria for Earth Quake design of	IS 1893 & IS 4326
32	Standard for CI double flanged pipe	IS 1536 & IS 1538
33	Steel Pipes	IS 3589 : 2001

Annexure II

List of applicable IS Codes

Some of the important IS codes referred during execution of the works are as follows:

IS:3764-	Safety code for excavation work.
IS:3720-	Methods of tests for soils, Concrete work.
IS:280-	Mild steel wire for general engineering purposes.
IS:269/8112-	Portland cement/OPC– grade43
IS:383-	Coarse and fine aggregates.
IS:432-	Medium tension steel bars and hard drawn steel wire.
IS1786-	Steel
IS:456-	Code of practice for Plain and reinforced concrete.
IS:516-	Methods of testing for strength of concrete
IS:1199 -	Method of sampling and analysis of concrete
IS:1786-	High strength deformed bars for Concrete reinforcement
IS:2386-	Methods of tests for aggregates of concrete.
IS:2506-	Code of Practice for bending and fixing of bars for concrete reinforcement